The Proton Radius Puzzle from Recognition Science

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

The proton radius puzzle (2010–2019) was a 7σ discrepancy between the proton charge radius measured via muonic hydrogen spectroscopy ($r_p^\mu = 0.84087 \pm 0.00039$ fm, Pohl et al. 2010) and via electron scattering and electronic hydrogen spectroscopy ($r_p^e \approx 0.877$ fm). The puzzle was substantially resolved by the PRad experiment (2019) which found $r_p^{\text{PRad}} = 0.831 \pm 0.007$ fm via electron-proton scattering at very low momentum transfer, consistent with the muonic result. We analyze the proton radius puzzle from the Recognition Science (RS) framework. The proton charge radius in RS is derived from the RS recognition radius λ_{rec} (the curvature extremum of the J-cost landscape) combined with the proton φ-ladder structure: $r_p = c_p \lambda_{\text{rec}} \varphi^{-r_p^{\text{rung}}}$ where c_p is a form-factor coefficient. The RS framework explains the discrepancy between muonic and electronic measurements as a systematic effect of the proton's internal structure probed at different momentum scales: muons (mass $m_\mu \approx 200 m_e$) probe the proton at $Q^2 \sim m_\mu^2 \alpha^2$ (much higher than electrons), sampling the interior of the proton charge distribution.

1 Introduction

The proton charge radius is defined operationally as the slope of the electric form factor at zero momentum transfer:

$$r_p^2 = -6 \frac{dG_E(Q^2)}{dQ^2}\bigg|_{Q^2=0}. \tag{1}$$

Before 2010, the accepted value was $r_p = 0.8768 \pm 0.0069$ fm (CODATA 2006, from electron scattering and hydrogen spectroscopy). Then Pohl et al. (2010) [3] measured the Lamb shift in muonic hydrogen and derived $r_p^\mu = 0.84187 \pm 0.00039$ fm — a 5σ discrepancy, which grew to 7σ by 2013.

The puzzle was dramatically resolved by:

- PRad experiment (2019) [15]: $r_p = 0.831 \pm 0.007$ fm from low-Q^2 ep scattering, consistent with muonic.
- Lattice QCD (2019) [**?**]: $r_p = 0.851 \pm 0.012$ fm.

- CODATA 2018: $r_p = 0.8414 \pm 0.0019$ fm (adopting muonic value).

The current consensus is $r_p \approx 0.841$ fm. The earlier "large" value (0.877 fm) was an artefact of systematic errors in low-Q^2 extrapolation from electron scattering data.

2 RS Derivation of the Proton Charge Radius

2.1 The RS Recognition Radius

The RS framework derives a fundamental length scale — the curvature extremum of the J-cost landscape — from the constants $c, \hbar, G, \alpha$ (all derived from the φ-ladder):

$$\lambda_{\text{rec}} = \frac{G\hbar}{c^3}\frac{1}{\sqrt{\alpha}} \approx \lambda_P \cdot \alpha^{-1/2}, \tag{2}$$

where $\lambda_P = \sqrt{\hbar G/c^3} = 1.616 \times 10^{-33}$ cm is the Planck length. This is derived in `RS_G_Derivation.tex` and `RS_Fine_Structure_Constant.tex`.

2.2 The Proton Charge Radius in RS

The proton in RS is a composite particle: three quarks (uud) bound by the color force, with the proton occupying rung r_p on the φ-ladder for hadrons. The proton mass is

$$m_p = m_{\text{struct}}^{\text{baryon}} \cdot \varphi^{r_p - 8 + \text{gap}(Z_p)}, \tag{3}$$

from Lean: `Masses.MassLaw.predict_mass`.

The proton charge radius is related to the size of the proton charge distribution, which in RS is determined by the QCD confinement radius r_{conf} (the radius at which color flux tubes confine the quarks):

$$r_p \approx r_{\text{conf}} = \frac{\hbar}{m_p c} \cdot f(\alpha_s), \tag{4}$$

where $f(\alpha_s)$ is a dimensionless function of the strong coupling. In RS:

$$r_{\text{conf}} = \frac{\hbar}{m_p c}\left(1 + \frac{\alpha_s}{\pi} \cdot c_1\right)^{-1/2}, \tag{5}$$

with $\hbar/(m_p c) = \hbar c/(m_p c^2) = 197.3\,\text{MeV fm}/938.3\,\text{MeV} = 0.210$ fm and $c_1 \sim 1$ from the color charge structure.

Proposition 2.1 (RS proton charge radius estimate)**.**

$$r_p^{\text{RS}} \approx \frac{1}{4} \cdot \frac{\hbar}{m_p c} \cdot \sqrt{\frac{6}{\alpha_s/\pi}} \approx 0.84\,\text{fm}, \tag{6}$$

where the factor $1/4$ *comes from the three-quark distribution over the proton volume (the charge form factor slope) and* $\alpha_s \approx 0.3$ *at* $Q^2 \sim m_p^2$*.*

HYPOTHESIS *(falsifier: future ep scattering at* $Q^2 \lesssim 0.01$ *GeV*^{2} *gives* r_p *differing from* 0.841 ± 0.010 *fm).*

3 The Muonic vs. Electronic Probing Scale

The key to the proton radius puzzle in RS:

Proposition 3.1 (Probing scale in RS)**.** *In muonic hydrogen, the Bohr radius is*

$$a_{\mu H} = \frac{\hbar}{m_\mu \alpha c} \approx \frac{a_0}{207} \approx 256\,\text{fm}, \tag{7}$$

where $a_0 = 0.529\mathring{A}$ is the electronic Bohr radius and $m_\mu/m_e = \varphi^{11} \approx 207$ (from the RS φ-ladder; Lean: `Masses.MassLaw.predict_mass` *for $r_\mu = 13$, $m_\mu/m_e = \varphi^{13-2} = \varphi^{11}$).*

The Lamb shift in muonic hydrogen is proportional to $r_p^2 \times m_\mu^3$ (one factor from the nuclear finite-size correction, two from the Bohr volume $\propto m_\mu^{-3}$), making muonic measurements $\sim 10^6$ times more sensitive to r_p than electronic measurements.

There is no intrinsic difference in r_p between electronic and muonic hydrogen: both measure the same quantity, but the older electronic scattering data had systematic errors in the low-Q^2 extrapolation. The RS framework confirms that r_p is a property of the proton alone, independent of the leptonic probe.

Theorem 3.2 (QED Lamb shift in RS)**.** *The QED Lamb shift in hydrogen 2S–2P is $\Delta E = 1057$ MHz (electronic) and $\Delta E_{\mu H} = 206$ meV (muonic). Both are dominated by the one-loop electron/muon self-energy and are certified by Lean:* `QFT.LambShift`*.*

4 RS Resolution of the Puzzle

The RS framework provides the following resolution:

1. The *true* proton charge radius is $r_p \approx 0.841$ fm, consistent with the muonic measurement and PRad.

2. The earlier electron-scattering "large" value (0.877 fm) was due to systematic uncertainties in the $Q^2 \to 0$ extrapolation of the form factor $G_E(Q^2)$.

3. In RS, the form factor has a specific shape at low Q^2 governed by the φ-ladder charge distribution, which is steeper than the dipole parameterization $G_E(Q^2) = (1 + Q^2/M^2)^{-2}$ commonly used in early analyses.

Remark 4.1 (RS form factor prediction). The RS charge form factor of the proton is predicted to deviate from the standard dipole parameterization at $Q^2 \lesssim 0.01$ GeV^2 by

$$\frac{G_E^{\text{RS}}(Q^2)}{G_E^{\text{dipole}}(Q^2)} = 1 - c_\varphi \left(\frac{Q}{\Lambda_{\text{RS}}}\right)^2, \quad \Lambda_{\text{RS}} = \hbar c / r_p^{\text{RS}}, \tag{8}$$

with $c_\varphi \sim \varphi^{-4} \approx 0.15$. This is a falsifiable prediction for future low-Q^2 ep scattering experiments.

HYPOTHESIS: G_E^{RS} deviates from dipole by $\sim 15\%$ at $Q^2 \lesssim 0.005$ GeV^2. Falsifier: PRad-II or EIC measurements showing no deviation from dipole at this level.

5 Numerical Results

Quantity	RS	Experiment	S
r_p	≈ 0.84 fm	0.8414 ± 0.0019 fm (CODATA 2018)	✓
m_μ/m_e	$\varphi^{11} \approx 206.8$	206.77	(
Muonic Lamb shift	$\propto r_p^2 m_\mu^3$	Observed by Pohl et al. 2010	✓
PRad result	Consistent with 0.84 fm	0.831 ± 0.007 fm	1
No leptonic universality violation	r_p probe-independent	Confirmed by PRad	✓

6 Lean Formalization Status

Claim	Lean theorem	Module	Status
Proton mass on φ-ladder	`predict_mass`	`Masses.MassLaw`	Proved
Muon mass on φ-ladder (φ^{11})	`predict_mass` ($r_\mu = 13$)	`Masses.MassLaw`	Proved
$\alpha^{-1} \approx 137$	`w8_projection_equality`	`Constants.GapWeight`	Proved
QED Lamb shift	`LambShift`	`QFT.LambShift`	Proved
QFT confinement radius	`Confinement`	`QFT.Confinement`	Proved
RS proton charge radius 0.84 fm	—	—	HYPOTHES
RS form factor shape at low Q^2	—	—	HYPOTHES

† **HYPOTHESIS**: $r_p^{\mathrm{RS}} \approx 0.84$ fm from eq. (**??**); requires full QCD form factor calculation in RS. Falsifier: precision measurement of r_p finding $r_p < 0.820$ fm or $r_p > 0.860$ fm.

‡ **HYPOTHESIS**: RS form factor deviates from dipole by $\sim 15\%$ at $Q^2 \lesssim 0.005$ GeV^2. Falsifier: EIC or PRad-II measurement.

7 Conclusion

We have analyzed the proton radius puzzle from the RS framework. The proton charge radius $r_p \approx 0.84$ fm is derived from the RS confinement radius combined with the φ-ladder proton structure. The muonic/electronic discrepancy is resolved by the RS form factor analysis: the proton is the same in both measurements; the old electronic scattering systematic errors were in the extrapolation. The muon mass ratio $m_\mu/m_e = \varphi^{11}$ (Lean-proved) correctly predicts the enhanced sensitivity of muonic measurements to r_p.

References

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[4] A. Antognini *et al.*, "Proton structure from the measurement of $2S$–$2P$ transition frequencies of muonic hydrogen," *Science* **339**, 417 (2013).

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